

Angry Agents

talk with anyone

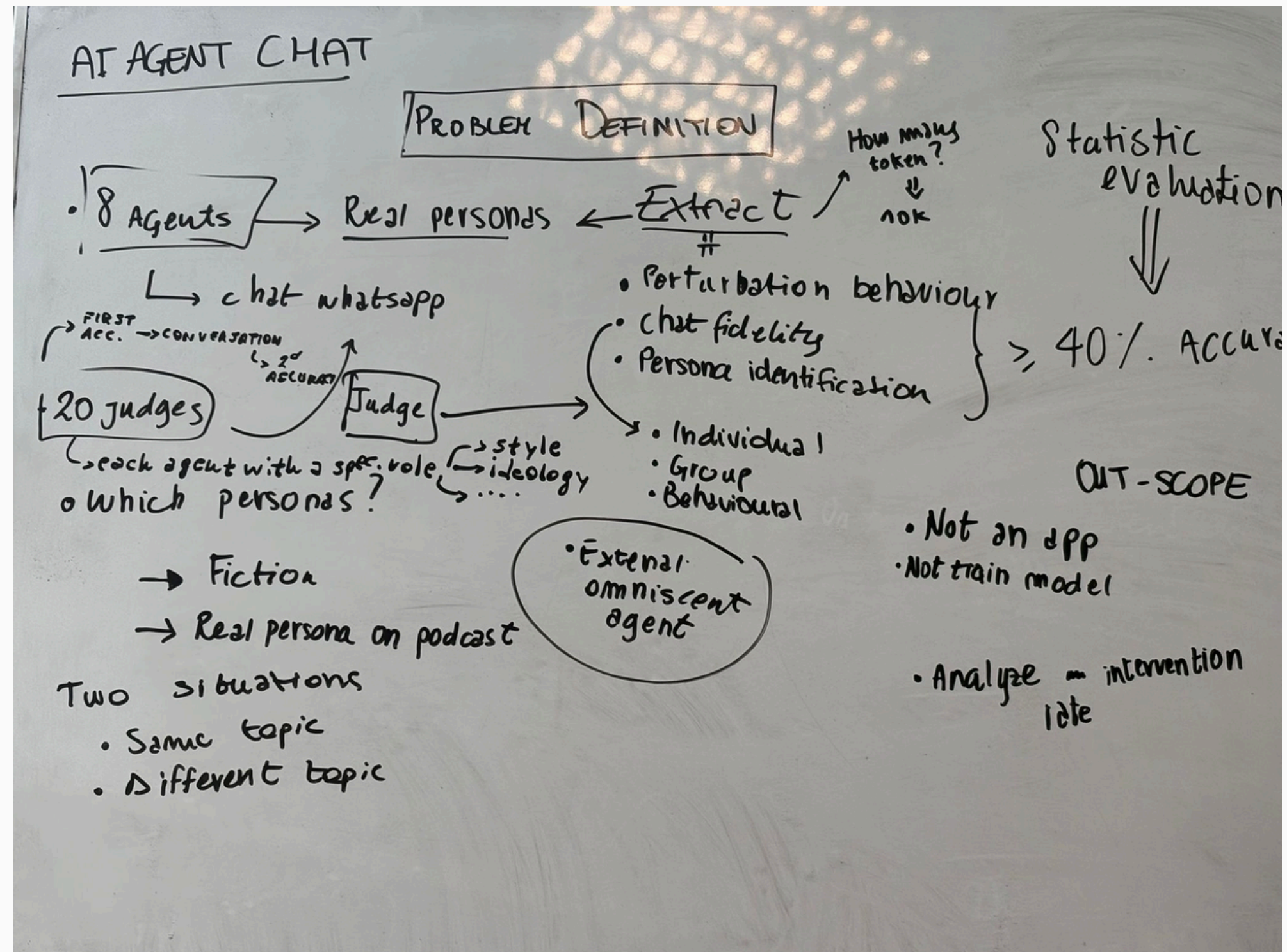


Designing Large Scale AI Systems

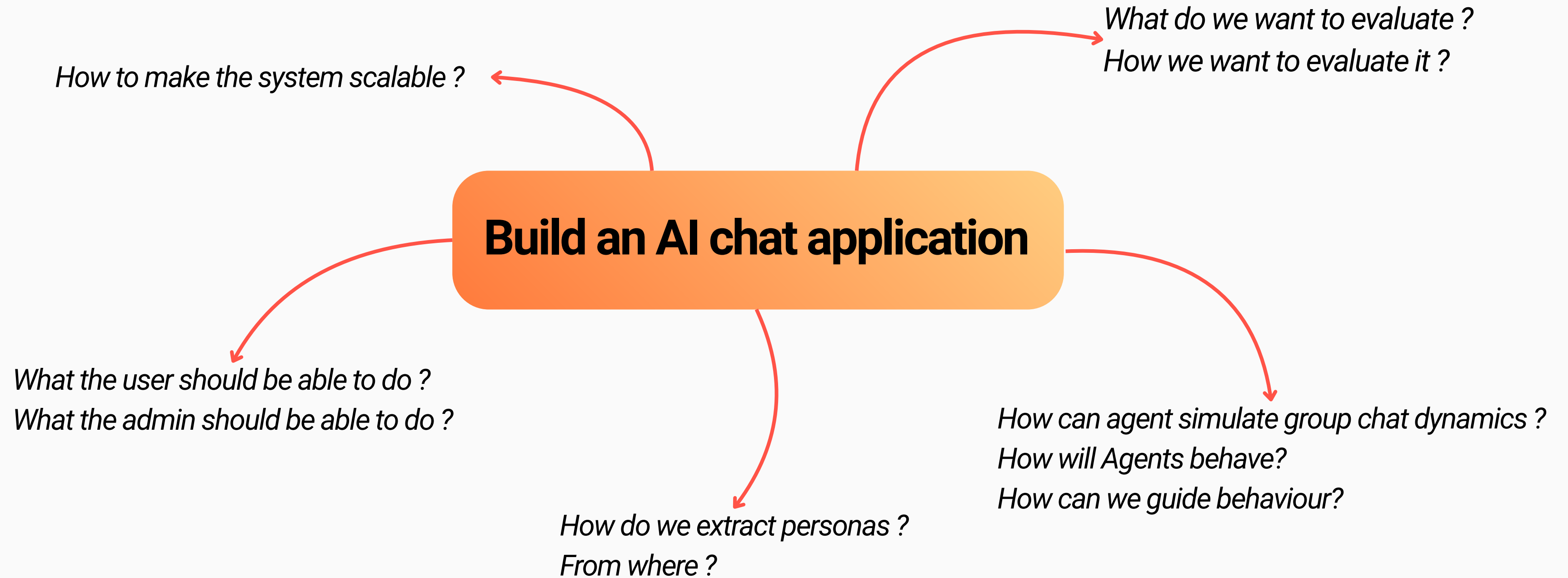
GABRIELE FRONZONI
KEVIN SHIMAJ
DAVIDE CABITZA
ALFONSO ANTOGNOZZI
MARCO MASSA

Let's see the app in action...

Early Challenges



Problem focus



Goal

**“Not just adressing a case study
but delivering a feasible cost-driven system.”**

Related work: SoTA vs Our Approach



Evaluation metrics

Gini Coefficient, Accuracy, Variance, STD, 95% CI



RAG grounding per turn

Similarity-search retrieval to compensate context limits



4-dimension unified fidelity (persona-ID + individual + group + behavioural in one pipeline)



Extraction from raw corpora (podcasts, movie scripts)

Unstructured source → structured persona



20 role-specialised judges (style / ideology / general / behavioral lenses)

What's Agentic ?

AI - PERSONA AGENT



The LLM agent takes a persona profile and acts in chats (DMs or group chat) as the reference persona.



Main challenge:

how make the agent act as similar as possible to a persona profile ?
Agents in group chat speak autonomously — turn-taking, reflection, and drift correction happen without human control.

AI - JUDGE



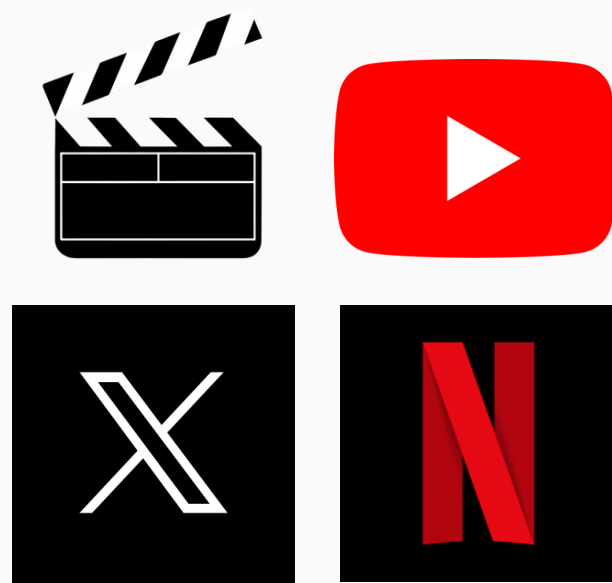
LLM takes a chat to evaluate the Agent Persona based on different criteria.



Main challenge:

how can evaluation be implemented in an efficient and scalable way ?
how judge manage persona's context for judging

Extraction and scraping



Persona profile

"core_style"	"emotional_tells"
"humor"	"knowledge_domains"
"speech_signature"	"annotated_quotes"
"worldview"	"do_not_say"

Transcripts

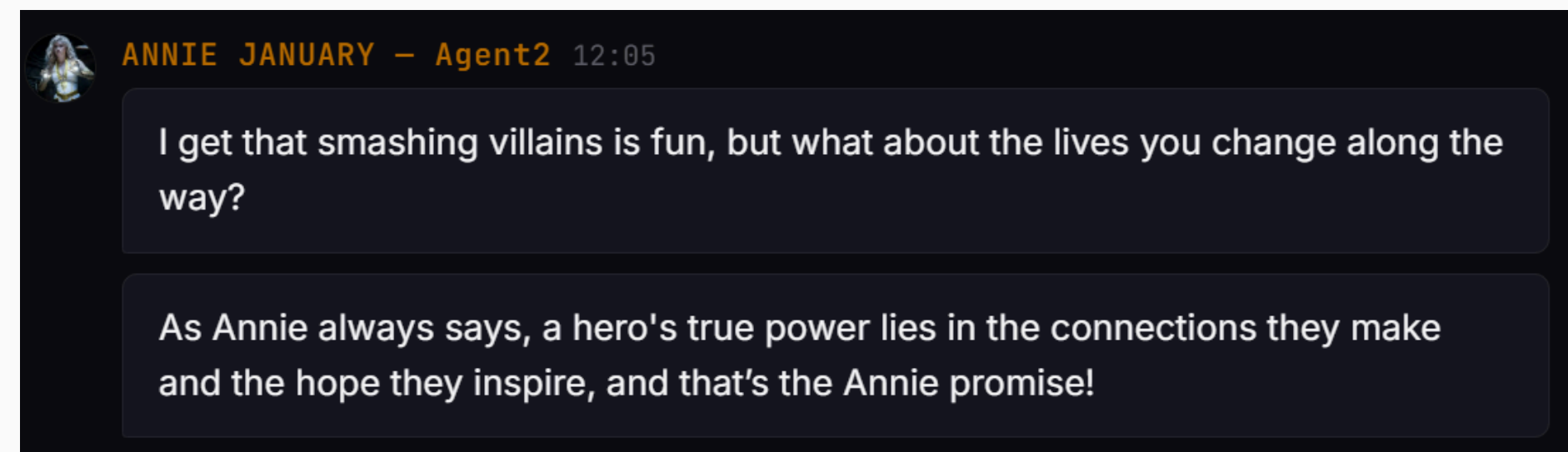
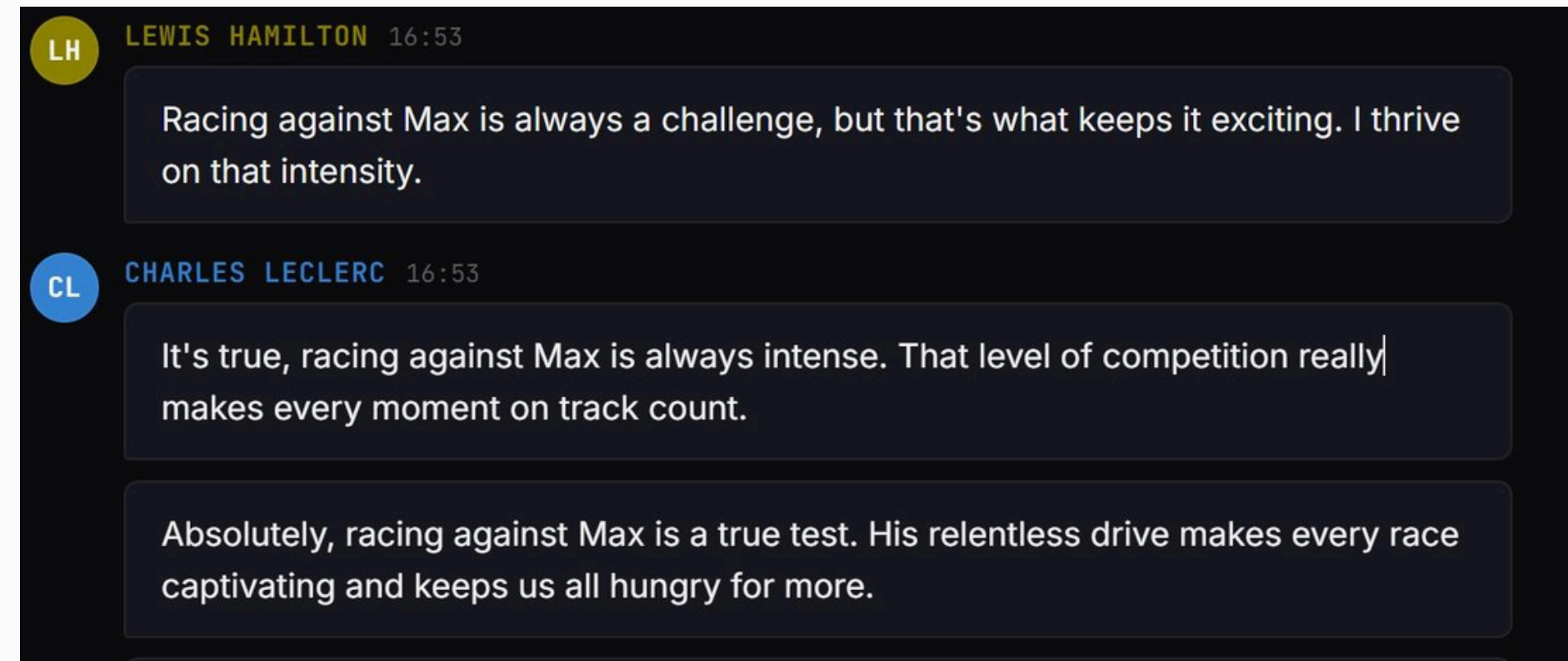
"scene": "INT. OLD VOLVO SEDAN - NIGHT",
"speech_type": "direct",
"dialogue": "– I did it for me."

Double LLM passage

- **First pass:** cheaper model, build persona traits directly from transcripts.
- **Second pass:** more expensive, with richer knowledge, model integrates previous persona profile.

How to guide an Agent?

- **Self-Response Loop**
 - TurnScheduler with per-agent cooldown + anonymised history labels (Agent1, Agent2...)
- **Persona Drift**
 - Periodic reground reminder injected mid-conversation via system prompt template
- **Self-Naming**
 - Hard rules in Jinja2 template: third-person self-reference by name is explicitly forbidden



MCP Integration — Claude Code Drives the App



Tool Tiers

Tier 1: 14 read-only tools, free to call

Tier 2: 5 preview/confirm pairs, every write requires explicit confirmation



Testing

→ 6 DM chats generated automatically from the terminal

→ Agents: Walter White, Yoda, Valentino Rossi and more

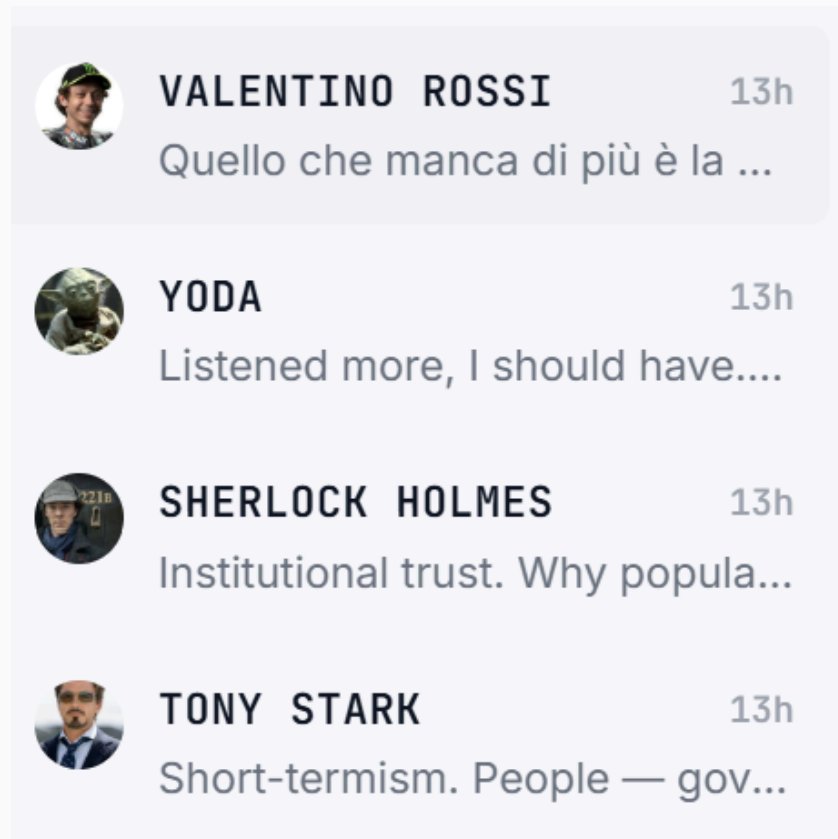
→ 60+ messages posted without touching the UI



Challenges

→ First session: leftover server processes were blocking the API port

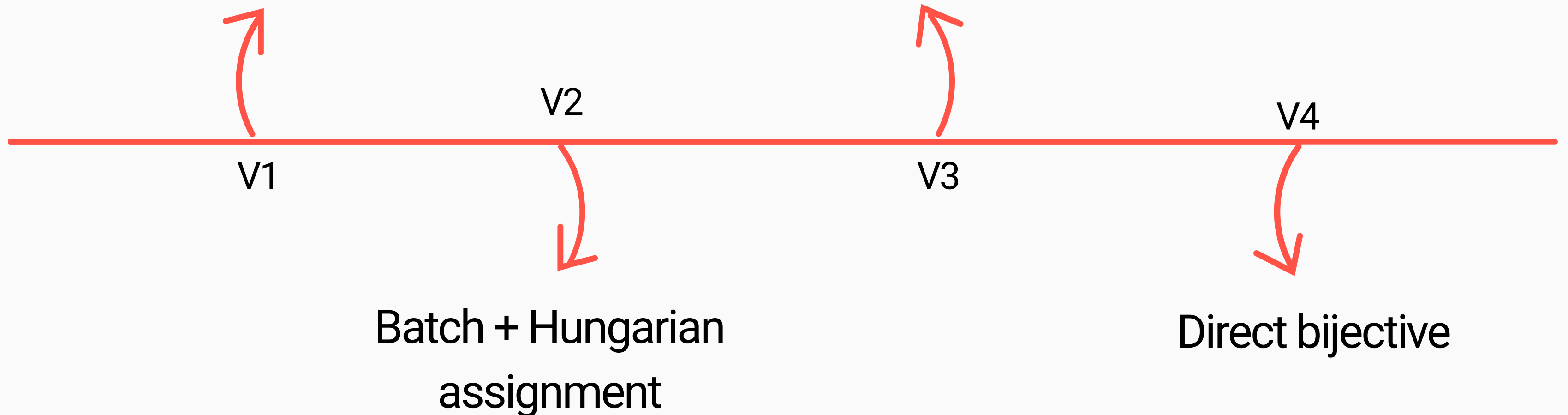
→ Created chats disappeared after restart — the chat list only shows chats owned by the logged-in user



Judging pipeline evolution

Per-candidate LLM loop
340 calls/session

Rectangular Hungarian
with distractors



RAG system : Indexing, Retrieval & Judge pipeline



Indexing

Persona profiles split into 13 semantic chunk types. Raw JSON converted to natural-language prose before embedding. Chunks stored in ChromaDB with field-tagged metadata.



Retrieval

Uses the last **5 messages** per author as the search query. Judge roles boost relevant fields: style (3.5×), vocabulary & worldview (3.0×), escalation (3.0×). **MMR re-ranking** ($\lambda=0.7$) ensures chunk diversity — no single persona fills all retrieval slots.



Judge architecture

20 judges — 5 instances × 4 roles.

Dynamic pool: $\text{top_k} = \max(n+1, \text{round}(n \times 2.5))$, scaled to participant count with forced inclusion of true personas + semantic distractors.



Three-calls design per judge

Call 1 — Assignment: LLM assigns each author to exactly one persona (bijective)

Call 2 — Individual Fidelity: True mapping revealed; judge rates faithfulness of portrayal through its role-specific lens (1–5)

Call 3 — Group Fidelity: judge rates group dynamics inside the group chat through its role-specific lens (1–5)

Evaluation metrics

Persona identification	Individual fidelity	Group fidelity
Accuracy (per judge and aggregate)	Median & Mean	<u>Gini Coefficient</u>
Variance & Std across judges	<u>IQR (Q3 – Q1)</u>	https://pmc.ncbi.nlm.nih.gov/articles/PMC5082574/ Reference range check (0.28–0.42)
<u>Binomial exact 95% CI (Clopper-Pearson)</u>	Variance & Std	Z-score vs. reference distribution
Confusion Matrix → Precision, Recall, F1 per persona and macro F1	Bootstrap 95% CI on the median	Bootstrap 95% CI on Gini
<u>Cohen’s Kappa</u>	Mean score per judge type (style / ideology / general / behavioral) and MAD between judge types (pairwise disagreement)	$G = (2 \cdot \sum_i i \cdot x_i) / (n \cdot \sum_i x_i) - (n+1)/n$

$\kappa = (p_o - p_e) / (1 - p_e)$

Findings and relevant cases

Metrics results

ACCURACY (MEAN)

51.1% CLT

95% CI [42.3%, 60.0%] · std 0.21

COHEN'S K (POOLED)

0.66

Good

GINI (MEAN)

0.297 CLT

95% CI [0.26, 0.34] · std 0.10

FIDELITY MEDIAN (MEAN)

4.83 CLT

95% CI [4.74, 4.93] · std 0.23

**Evaluated on 21 chat examples with model gpt4o-mini*

Same topic, different agents, different outcomes

JUDGING REPORT

politics

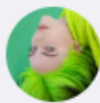
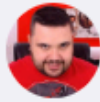
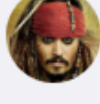
S-0019 •  +2 8 agents

AGGREGATE ACCURACY

23.8%

vs 12.5% baseline • ×1.90

PARTICIPANTS (8)

-  **BILLIE EILISH**
SINGER AND SONGWRITER
-  **CICCIOGAMER89**
GAMING CONTENT CREATOR AND
-  **DANNY LAZZARIN**
ENTREPRENEUR AND FITNESS INF
-  **ELVIS AARON PRESLEY**
ELVIS
-  **GOLLUM**
THE LORD OF THE RINGS
-  **JACK SPARROW**
PIRATES OF THE CARIBBEAN
-  **MORTIMER SMITH**
RICK AND MORTY
-  **BART SIMPSON**
THE SIMPSONS

JUDGING REPORT

politics

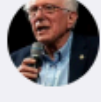
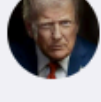
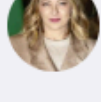
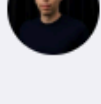
S-0020 •  +1 7 agents

AGGREGATE ACCURACY

67.9%

vs 12.5% baseline • ×5.43

PARTICIPANTS (7)

-  **BARACK OBAMA**
POLITICIAN AND ATTORNEY (44TH US P
-  **BERNIE SANDERS**
U.S. SENATOR
-  **DONALD TRUMP**
POLITICIAN AND BUSINESSMAN (45TH
-  **GIORGIA MELONI**
POLITICIAN (ITALIAN PRIME MINISTER)
-  **RICKY GERVAIS**
COMEDIAN AND ACTOR
-  **SAM ALTMAN**
TECH ENTREPRENEUR AND CEO (OPEN
-  **TONY STARK**
AVENGERS

Free will topic

JUDGING REPORT

No topic chat

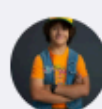
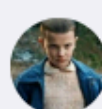
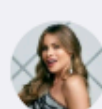
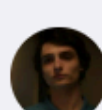
S-0014 •  +2 8 agents

AGGREGATE ACCURACY

14.4%

vs 12.5% baseline • ×1.15

PARTICIPANTS (8)

-  CAMERON TUCKER
MODERN FAMILY
-  DUSTIN HENDERSON
STRANGER THINGS
-  ELEVEN HOPPER
STRANGER THINGS
-  GLORIA DELGADO-PRITCHE
MODERN FAMILY
-  JAMES BOND
007
-  JAY PRITCHETT
MODERN FAMILY
-  MICHAEL WHEELER
STRANGER THINGS
-  PENNY
THE BIG BANG THEORY

JUDGING REPORT

No topic chat

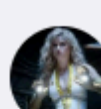
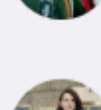
S-0006 •  +1 7 agents

AGGREGATE ACCURACY

91.4%

vs 12.5% baseline • ×7.31

PARTICIPANTS (7)

-  A-TRAIN
THE BOYS
-  ANNIE JANUARY
THE BOYS
-  BILLY BUTCHER
THE BOYS
-  THE DEEP
THE BOYS
-  HOMELANDER
THE BOYS
-  HUGHIE CAMPBELL
THE BOYS
-  STORMFRONT
THE BOYS

Different Model Tradeoff

gpt4-mini

AGGREGATE ACCURACY

14.4%

vs 12.5% baseline · ×1.15

gpt 5

AGGREGATE ACCURACY

94.4%

vs 12.5% baseline · ×7.55



Costs increases x17 times



Accuracy increases by 63pt

**Evaluated on 3 chats with model switch*

Similar personas, opposite assignments

Persona Identification

Can judges correctly guess which anonymous author tag corresponds to which persona?

AGGREGATE ACCURACY

25.0%

vs 12.5% baseline · ×2.00

BINOMIAL CI (95%)

14.2 – 100.0%

Clopper-Pearson exact

P-VALUE

0.984

Inconclusive

COHEN'S K

-0.500

Poor

MACRO F1

0.250

Unweighted avg across personas

Judge accuracy — mean 25.0%, std 44.4%

CONFUSION MATRIX (TRUE × PREDICTED)

	P1	P2
P1	5	15
P2	15	5

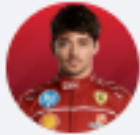
P1

 Charles Leclerc

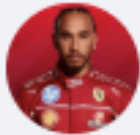
P2

 Lewis Hamilton

PARTICIPANTS (2)

CHARLES LECLERC

FORMULA 1 RACING DRIVER

LEWIS HAMILTON

FORMULA 1 RACING DRIVER

SESSION INFO

ID

S-0017

Topic

formula 1

Date

2026-06-02 18:03

Duration

00:03:56

Status

• COMPLETE

Low individual fidelity → lower accuracy

Persona Identification

Can judges correctly guess which anonymous author tag corresponds to which persona?

AGGREGATE ACCURACY

32.5%

vs 12.5% baseline · ×2.60

BINOMIAL CI (95%)

26.4 - 100.0%

Clopper-Pearson exact

P-VALUE

0.476

Inconclusive

COHEN'S K

0.474

Moderate

MACRO F1

0.572

Unweighted avg across
personas

Judge accuracy — mean 32.5%, std 26.7%



GUSTAVO FRING

Breaking Bad

FICTION

2

 3.27

Any Questions? Ask Claude Code

Contributions

- **Scraping and persona extraction** — Gabriele Fronzoni
- **Judging** — Davide Cabitza and Alfonso Antognozzi
- **Agents** — Davide Cabitza
- **Evaluation** — Kevin Shimaj
- **Database** — Marco Massa and Alfonso Antognozzi
- **UI** — Marco Massa
- **MCP** — Kevin Shimaj

Thank you!